Question 1

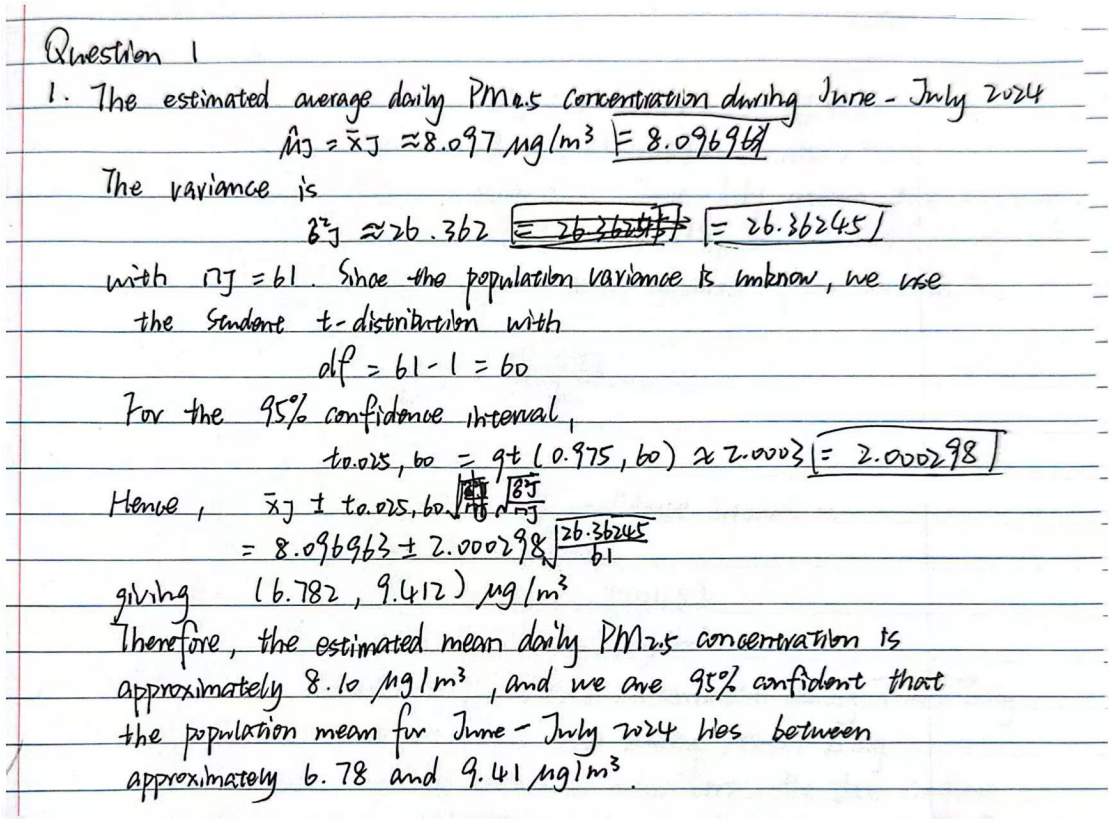
1.1

```r
# 1.1
june_july <- read.csv("pm25.june_july_2024.csv")

mean(june_july$pm25)
var(june_july$pm25)
length(june_july$pm25)

qt(1- 0.05/2, df = 60)
```

```
> mean(june_july$pm25)
[1] 8.096963
> var(june_july$pm25)
[1] 26.36245
> length(june_july$pm25)
[1] 61
>
> qt(1- 0.05/2, df = 60)
[1] 2.000298
```



Question 1

1. The estimated average daily $PM_{2.5}$ concentration during June - July 2024

$$\hat{\mu}_J = \bar{x}_J \approx 8.097\ \mu g/m^3 \quad [= 8.096963]$$

The variance is

$$\hat{\sigma}^2_J \approx 26.362 \quad [= 26.36245]$$

with $n_J = 61$. Since the population variance is unknown, we use the Student t-distribution with

$$df = 61 - 1 = 60$$

For the 95% confidence interval,

$$t_{0.025,60} = qt(0.975, 60) \approx 2.0003 \quad [= 2.000298]$$

Hence, $\bar{x}_J \pm t_{0.025,60}\sqrt{\frac{\hat{\sigma}^2_J}{n_J}}$

$$= 8.096963 \pm 2.000298\sqrt{\frac{26.36245}{61}}$$

giving $(6.782, 9.412)\ \mu g/m^3$

Therefore, the estimated mean daily $PM_{2.5}$ concentration is approximately $8.10\ \mu g/m^3$, and we are 95% confident that the population mean for June - July 2024 lies between approximately 6.78 and $9.41\ \mu g/m^3$.

1.2

```r
# 1.2
march_may <- read.csv("pm25.march_may_2024.csv")

mean(march_may$pm25)
var(march_may$pm25)
length(march_may$pm25)
```

```
> mean(march_may$pm25)
[1] 10.3629
> var(march_may$pm25)
[1] 35.48596
> length(march_may$pm25)
[1] 84
```

2. The sample:

$\bar{x}_J = 8.096963 \quad \hat{\sigma}^2_J = 26.36245 \quad n_J = 61$

$\bar{x}_M = 10.3629 \quad \hat{\sigma}^2_M = 35.48596 \quad n_M = 84$

The estimated difference, defined as June–July minus March–May,

$\hat{\mu}_J - \hat{\mu}_M = 8.096963 - 10.3629 = -2.265937 \ \mu g/m^3$

Use the approximate method unknown variance from Lecture 4,

$$SE = \sqrt{\frac{\hat{\sigma}^2_J}{n_J} + \frac{\hat{\sigma}^2_M}{n_M}} = \sqrt{\frac{26.36245}{61} + \frac{35.48596}{84}}$$

$$= 0.92446$$

Therefore, the approximate 95% confidence interval is

$$(\hat{\mu}_J - \hat{\mu}_M) \pm z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{\sigma}^2_J}{n_J} + \frac{\hat{\sigma}^2_M}{n_M}}$$

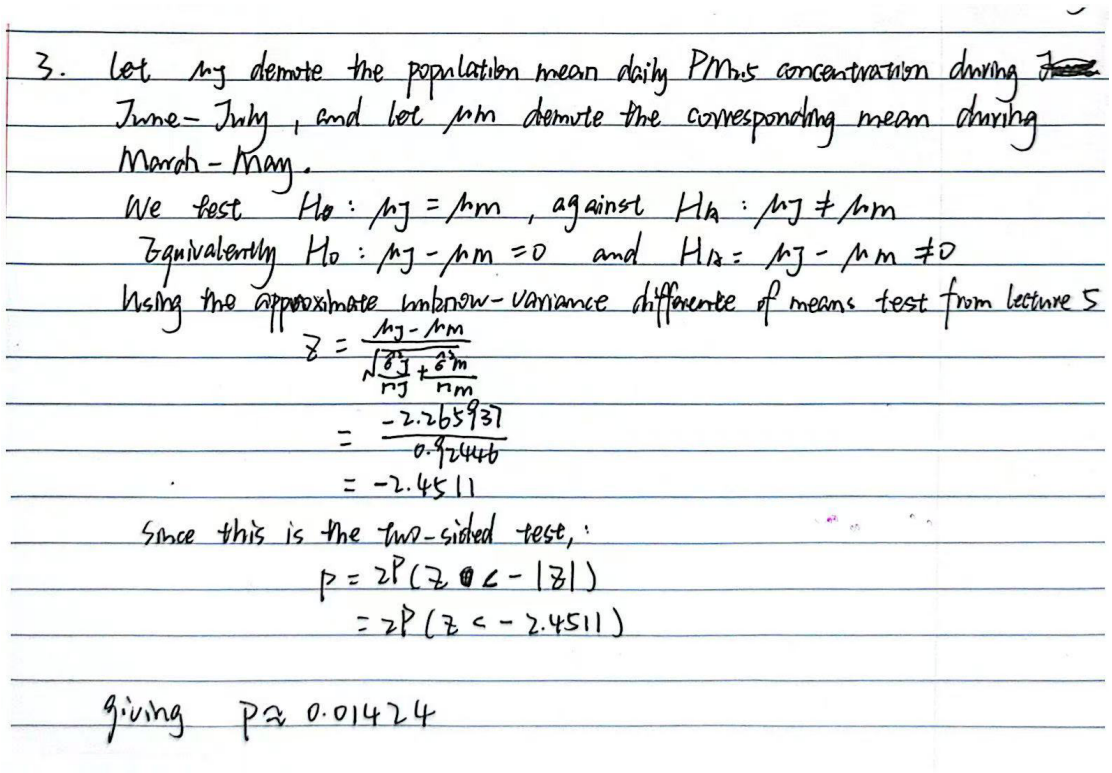
$$= -2.265937 \pm 1.96(0.92446)$$

giving $(-4.078, -0.454) \ \mu g/m^3$

Hence, the estimated mean $PM_{2.5}$ concentration during June–July was about $2.27 \ \mu g/m^3$ lower than during March–May.

Since the entire confidence interval is below zero, the data suggest that the population mean $PM_{2.5}$ concentration was lower during June–July.

1.3




```
# 1.3
2 * pnorm(-2.4511)
```

```
> # 1.3
> 2 * pnorm(-2.4511)
[1] 0.01424204
```

This means that if the two population means were truly equal, the probability of observing a standardized difference at least as extreme as the one observed is approximately 1.4%.

Since p < 0.05, there is evidence against the null hypothesis at the 5% significance level. Therefore, the data suggest that the population mean daily PM2.5 concentration differed between the two periods. Since the estimated difference (June-July minus March-May) is negative, the June-July mean appears to be lower.

1.4

```r
# 1.4
march_may$x <- 0
june_july$x <- 1

pm25data <- rbind(march_may, june_july)

model <- lm(pm25 ~ x, data = pm25data)

summary(model)
```

```
Call:
lm(formula = pm25 ~ x, data = pm25data)

Residuals:
   Min     1Q Median     3Q    Max
-6.285 -3.920 -1.427  1.810 21.085

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  10.3629     0.6139  16.880   <2e-16 ***
x            -2.2659     0.9465  -2.394    0.018 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.627 on 143 degrees of freedom
Multiple R-squared:  0.03853,   Adjusted R-squared:  0.03181
F-statistic: 5.731 on 1 and 143 DF,  p-value: 0.01796
```

4. Define $x = \begin{cases} 0, & \text{March-May} \\ 1, & \text{June-July} \end{cases}$

The estimated coefficient of x is,

$$\hat{\beta}_1 = -2.26594$$

In the linear model,

$$E[PM_{2.5}|x] = \beta_0 + \beta_1 x$$

β_0 represents the mean for the $x=0$ group, March-May while $\beta_0 + \beta_1$ represents the mean for the $x=1$ group June-July.

Therefore, $\hat{\beta}_1 = \bar{x}_J - \bar{x}_M = -2.26594$

Hence, the estimated coefficient of x is exactly the same as the estimated difference in means obtained in Question 1.2.

Question2

Question 2.

1. $p(y|v) = \exp(-v)\exp[-\exp(-v)y]$

$= e^{-v} e^{-e^{-v}y}$

likelihood,

$p(y|v) = \prod_{i=1}^{n} p(y_i|v)$

$= \prod_{i=1}^{n} [e^{-v} e^{-e^{-v}y_i}]$

$= \prod_{i=1}^{n} e^{-v} \prod_{i=1}^{n} e^{-e^{-v}y_i}$

$= [e^{-v}]^n e^{-e^{-v}y_1} e^{-e^{-v}y_2} + \cdots + e^{-e^{-v}y_n}$

$= e^{-nv} e^{-e^{-v}(y_1+y_2+\cdots y_n)}$

$= e^{-nv} e^{-e^{-v}\sum_{i=1}^{n} y_i}$

$= \exp(-nv - e^{-v}\sum_{i=1}^{n} y_i)$

2. negative log-likelihood,

$L(y|v) = -\log p(y|v)$

$= -\log[\exp(-nv - e^{-v}\sum_{i=1}^{n} y_i)]$

$= nv + e^{-v}\sum_{i=1}^{n} y_i$

3. $L(y|v) = nv + e^{-v}\sum_{i=1}^{n} y_i$

Differentiating with respect to v,

$\frac{dL}{dv} = n - e^{-v}\sum_{i=1}^{n} y_i$

Setting the derivative $= 0$

$n - e^{-v}\sum_{i=1}^{n} y_i = 0$

$n = e^{-v}\sum_{i=1}^{n} y_i$

$e^{-v} = \frac{n}{\sum_{i=1}^{n} y_i}$

using the sample mean

$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$

$\sum_{i=1}^{n} y_i = n\bar{y}$

so, $e^{-\hat{v}} = \frac{n}{n\bar{y}} = \frac{1}{\bar{y}}$

taking log gives, $-\hat{v} = -\log\bar{y}$

and hence, $\hat{v} = \log\bar{y}$

Also, $\frac{d^2L}{dv^2} = e^{-v}\sum_{i=1}^{n} y_i > 0$

so this value minimises the negative log-likelihood.

4. estimator becomes the random variable,

$$\hat{v} = \log \bar{Y}$$

where, $E[\bar{Y}] = e^v \qquad V[\bar{Y}] = \frac{e^{2v}}{n}$

Let $f(x) = \log x$

Then $f'(x) = \frac{1}{x} \qquad f''(x) = -\frac{1}{x^2}$

Using the lecture 2 approximation,

$$E[f(x)] \approx f(\mu_x) + \frac{\sigma_x^2}{2} f''(\mu_x).$$

$$E[\hat{v}] \approx f(e^v) + \frac{e^{2v}/n}{2} f''(e^v)$$

$$\approx \log(e^v) + \frac{e^{2v}/n}{2}\left(-\frac{1}{(e^v)^2}\right)$$

$$\approx v + \frac{e^{2v}/n}{2}\left(-\frac{1}{e^{2v}}\right)$$

$$\approx v + \left(-\frac{1}{2n}\right)$$

So, $E[\hat{v}] \approx v - \frac{1}{2n}$

Therefore, $\text{Bias}[\hat{v}] = E[\hat{v}] - v$

$$\approx v - \frac{1}{2n} - v$$

$$\approx -\frac{1}{2n}$$

$$\text{Bias}(\hat{v}) \approx -\frac{1}{2n}$$

Using the lecture 2 approximation,

$$V[f(x)] \approx \sigma_x^2 \times [f'(\mu_x)]^2$$

$$V[\hat{v}] \approx \frac{e^{2v}}{n}\left(\frac{1}{e^v}\right)^2$$

$$\approx \frac{e^{2v}}{n} \cdot \frac{1}{e^{2v}}$$

$$\approx \frac{1}{n}$$

Finally, $\text{Bias}(\hat{v}) \approx -\frac{1}{2n}$

$$V[\hat{v}] \approx \frac{1}{n}$$

2.5

5. From Question 2.4, the approximate bias of the maximum likelihood estimator is, $\text{Bias}(\hat{v}) \approx -\frac{1}{2n}$

Therefore, an approximately bias-corrected estimator is

$$\hat{v}_{BC} = \log \bar{Y} + \frac{1}{2n}$$

For $v = 1$ $n = 10$, the approximate bias of the original MLE is

$$-\frac{1}{2\times 10} = -0.05$$

```r
# 2.5
set.seed(1)

v <- 1
n <- 10
N <- 10000

# create two empty vectors of length 10000
vhat <- numeric(N)
vhat_bc <- numeric(N)

for (i in 1:N) {
  # generate 10 exponential observations
  y <- rexp(n, rate = exp(-v))

  vhat[i] <- log(mean(y)) # from 2.3

  vhat_bc[i] <- vhat[i] + 1 / (2 * n)
}

bias_vhat <- mean(vhat) - v
bias_vhat_bc <- mean(vhat_bc) - v

bias_vhat
bias_vhat_bc
```

```
> bias_vhat
[1] -0.04911784
> bias_vhat_bc
[1] 0.0008821578
```

The situation shows that the estimated bias of the original MLE is approximately -0.05, while the estimated bias of the bias-corrected estimator is close to 0. Thus, the bias correction appears to work well for n = 10, although a small remaining bias may occur because the bias formula itself is an approximation and the simulation is random.

Question 3

Question 3

1. $\hat{\theta}_A = \frac{44}{50} = 0.88$.

Using the CLT approximate confidence interval for a sample mean,

$$\hat{\theta}_A \pm 1.96\sqrt{\frac{\hat{\theta}_A(1-\hat{\theta}_A)}{n_A}}$$

Turn into

$$0.88 \pm 1.96\sqrt{\frac{0.88(0.12)}{50}}$$

Thus,

$$95\%\,CI = (0.790, 0.970)$$

Therefore, the estimated probability that a ~~pair~~ pair from Supplier A is construction grade is 0.88 (88%).

The approximate 95% confidence interval is approximately 79% to 97%.

2. $\hat{\theta}_B = \frac{36}{50} = 0.72$

$\hat{\theta}_A - \hat{\theta}_B = 0.88 - 0.72 = 0.16$

Using the CLT approximation,

$$SE = \sqrt{\frac{\hat{\theta}_A(1-\hat{\theta}_A)}{n_A} + \frac{\hat{\theta}_B(1-\hat{\theta}_B)}{n_B}}$$

$$= \sqrt{\frac{0.88(0.12)}{50} + \frac{0.72(0.28)}{50}}$$

$$= 0.07838$$

Hence, $0.16 \pm 1.96(0.07838)$

gives,

$$95\%\,CI = (0.0064, 0.3136)$$

Therefore, Supplier A's estimated construction grade probability is 0.16 higher than Supplier B's.

The approximate 95% confidence interval is entirely above 0, suggesting that Supplier A may have a higher construction grade probability.

3. $H_0 : \theta_A = \theta_B$

against $H_A : \theta_A \neq \theta_B$

The pooled estimate is,

$$\hat{\theta}_p = \frac{44+36}{50+50} = 0.80$$

Thus,

$$Z(\hat{\theta}_A - \hat{\theta}_B) = \frac{\hat{\theta}_A - \hat{\theta}_B}{\sqrt{\hat{\theta}_p(1-\hat{\theta}_p)(\frac{1}{n_A}+\frac{1}{n_B})}}$$

$$= \frac{0.88 - 0.72}{\sqrt{0.80(0.20)(\frac{1}{50}+\frac{1}{50})}}$$

$$= \frac{0.16}{0.08}$$

$$= 2$$

For the two-sided test,

$$p \approx 2P(Z < -2)$$

give giving $p \approx 0.0455$

Since $p < 0.05$, the null hypothesis is rejected at the 5% significance level. The data provide evidence that ~~at~~ the construction grade probabilities differ between the two suppliers, with Supplier A having the higher observed probability.

4. For M_0, all two pours are assumed to share a common Bernoulli probability p. There are 80 construction grade pours and 20 non-construction grade pours, so the likelihood is,

$$L_0(p) = p^{80}(1-p)^{20}$$

The negative log-likelihood is

$$NLL_0(p) = -[80\log(p) + 20\log(1-p)]$$

The maximum likelihood estimate is,

$$\hat{p} = \frac{80}{100} = 0.8$$

Therefore,

$$NLL_0 = -[80\log(0.8) + 20\log(0.2)]$$
$$= 50.0402$$

Model M_0 has one independently estimated parameter, so $k_0 = 1$

Therefore,

$$AIC_0 = NLL_0 + k_0$$
$$= 50.0402 + 1$$
$$= 51.0402$$

and, $$BIC_0 = NLL_0 + \frac{k_0}{2}\log n$$
$$= 50.0402 + \frac{1}{2}\log(100)$$
$$= 52.3428$$

For M_1, Supplier A and Supplier B have separate Bernoulli probabilities. Their maximum likelihood estimates are,

$$\hat{p}_A = \frac{44}{50} = 0.88$$

and, $$\hat{p}_B = \frac{36}{50} = 0.72$$

The likelihood is,

$$L_1(p_A, p_B) = p_A^{44}(1-p_A)^6 p_B^{36}(1-p_B)^{14}$$

Therefore, the negative log-likelihood is,

$$NLL_1 = -[44\log(0.88) + 6\log(0.12) + 36\log(0.72) + 14\log(0.28)]$$
$$= 47.9939$$

Model M_1 has two independently estimated parameters, so $k_1 = 2$.

Therefore,

$$AIC_1 = NLL_1 + k_1$$
$$= 47.9939 + 2$$
$$= 49.9939$$

and, $$BIC_1 = NLL_1 + \frac{k_1}{2}\log(n)$$
$$= 47.9939 + \frac{2}{2}\log(100)$$
$$= 52.5991$$

Since a smaller information criterion is preferred,

AIC select M_1

because $49.9939 < 51.0402$

Then, BIC select M_0

because $52.3428 < 52.5991$

3.5

The hypothesis test gives $p \approx 0.0455$, so at the 5% significance level there is evidence against equal construction grade probabilities. The AIC also selects M1, the model with separate probabilities for the two suppliers. However, the BIC selects the simpler model M0, in which both suppliers have the same probability.

Therefore, the three approaches do not all lead to the same conclusion. Overall, the results provide some evidence that the construction grade probability differ between Supplier A and Supplier B, but the evidence is relatively weak or borderline rather than overwhelming.